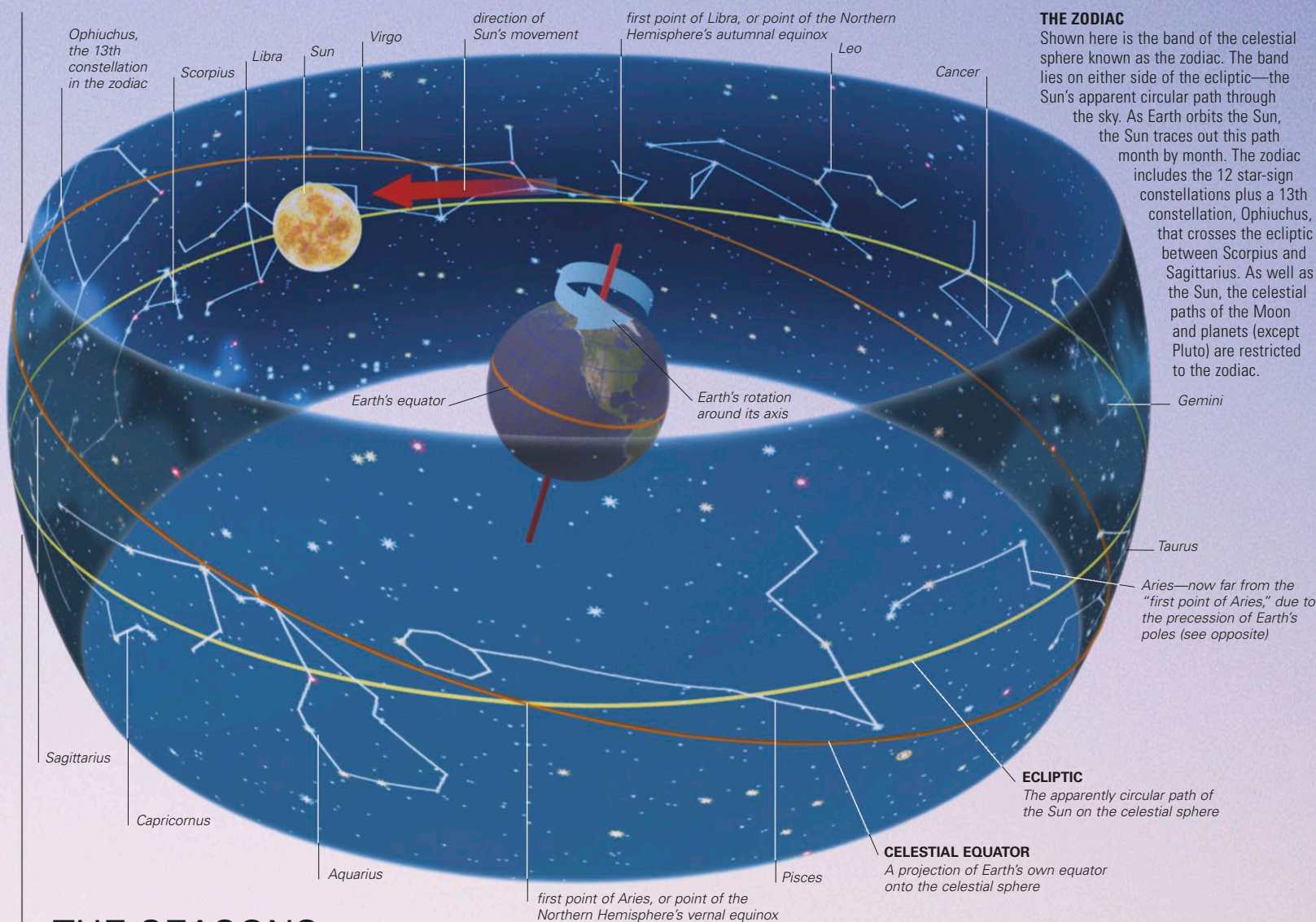




THE ZODIAC

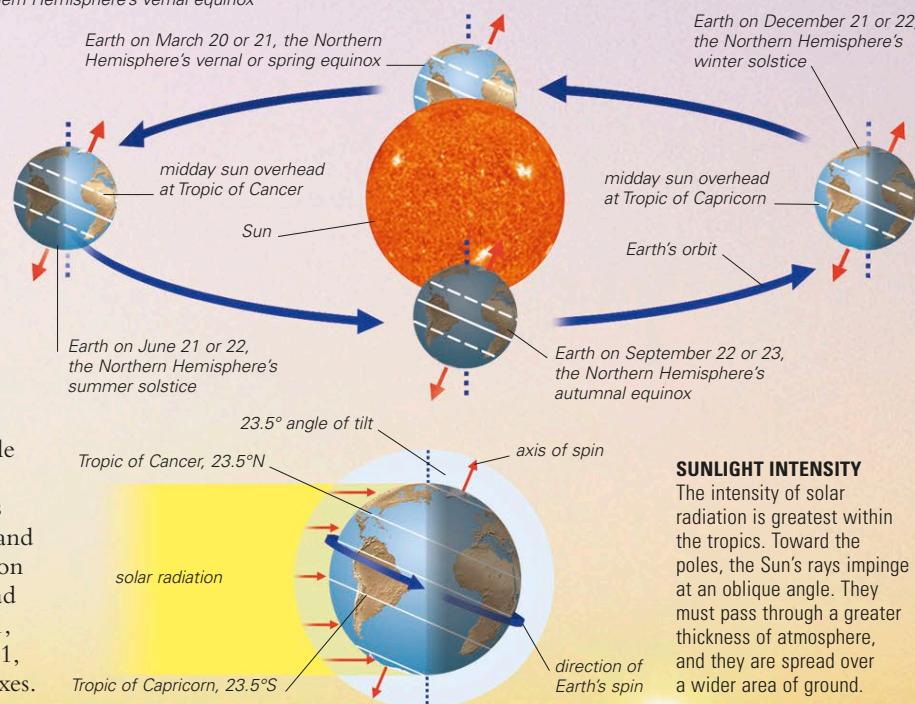
Shown here is the band of the celestial sphere known as the zodiac. The band lies on either side of the ecliptic—the Sun's apparent circular path through the sky. As Earth orbits the Sun, the Sun traces out this path month by month. The zodiac includes the 12 star-sign constellations plus a 13th constellation, Ophiuchus, that crosses the ecliptic between Scorpius and Sagittarius. As well as the Sun, the celestial paths of the Moon and planets (except Pluto) are restricted to the zodiac.

THE SEASONS

Earth's orbit around the Sun takes 365.25 days and provides a key unit of time, the year. Earth's seasons result from the tilt of its axis relative to its orbit. Due to Earth's tilt, one or other of its hemispheres is normally pointed toward the Sun. The hemisphere that tilts toward the Sun receives more sunlight and is therefore warmer. Each year, the Northern Hemisphere reaches its maximum tilt toward the Sun around June 21—summer solstice in the Northern Hemisphere and winter solstice in the Southern Hemisphere. For some time around this date, the north polar region is sunlit all day, while the south polar region is in darkness. Conversely, around December 21, the situation is reversed. Between the solstices are the equinoxes, when Earth's axis is broadside to the Sun and the periods of daylight and darkness are equal for all points on Earth. Earth's tilt also defines the tropics. The Sun is overhead at midday on the Tropic of Cancer (23.5°N) around June 21, above the Tropic of Capricorn (23.5°S) around December 21, and directly above the equator at midday during the equinoxes.

SOLSTICES AND EQUINOXES

At the solstices, in June and December, one hemisphere has its longest day, the other its shortest. At the equinoxes, in March and September, the lengths of day and night are equal everywhere on Earth.



SUNLIGHT INTENSITY

The intensity of solar radiation is greatest within the tropics. Toward the poles, the Sun's rays impinge at an oblique angle. They must pass through a greater thickness of atmosphere, and they are spread over a wider area of ground.